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Batch:52

Assignment-9.2

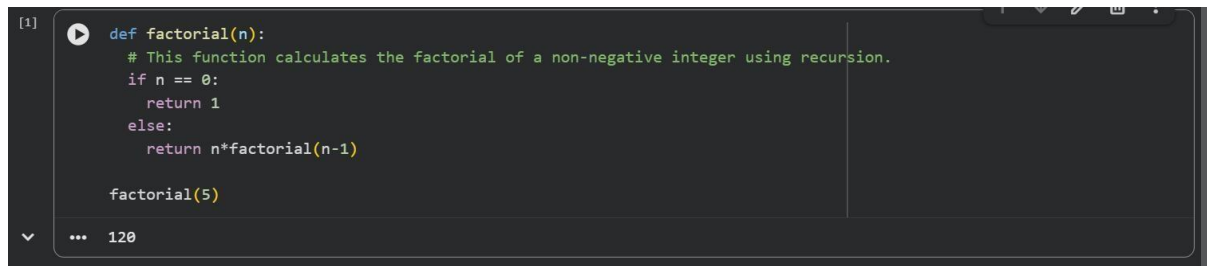
Task Description -1 (Documentation – Function Summary Generation)

Code used: def
factorial(n):
if n == 0:
 return 1
else:
 return n*factorial(n-1)
factorial(5)

Prompt:

Add a short summary describing the purpose of the function in the code

Output:



```
[1] def factorial(n):  
    # This function calculates the factorial of a non-negative integer using recursion.  
    if n == 0:  
        return 1  
    else:  
        return n*factorial(n-1)  
  
    factorial(5)
```

... 120

Task Description -2 (Documentation – Logical Explanation for Conditions and Loops) Code Used:

```
def numbers(n):  
    if  
        n > 0:  
            result = "Positive"  
        elif n < 0: result =  
            "Negative" else:  
            result = "Zero"  
    return result  
numbers(3)
```

Prompt: explain only decision-making logic and

loop behavior inside the code Output:

```
[7] def numbers(n):
    # Decision-making logic: Determines if 'n' is positive, negative, or zero.
    # No loops are present in this function; execution flows through conditional branches.
    # Check if 'n' is positive
    if n > 0:
        result = "Positive"
    # If not positive, check if 'n' is negative
    elif n < 0:
        result = "Negative"
    # If neither positive nor negative, 'n' must be zero
    else:
        result = "Zero"
    return result
numbers(3)

... 'Positive'
```

Task Description -3 (Documentation – File-Level Overview)

Code Used: def

factorial(n): result = 1 for

i in range(1, n + 1):

result *= i

return result

def square_list(numbers): return

[n * n for n in numbers]

def main(): print(factorial(5))

print(square_list([1, 2, 3, 4])) main()

Prompt:

Generate a high-level overview summarizing the overall purpose and functionality of the Code. add the summary at the beginning

Output:

```
[8] ✓ 0s # This code defines two utility functions: `factorial` to calculate the factorial of a number,
# and `square_list` to square each element in a list. It then demonstrates their usage in a `main` function.

def factorial(n):
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result

def square_list(numbers):
    return [n * n for n in numbers]

def main():
    print(factorial(5))
    print(square_list([1, 2, 3, 4]))

main()

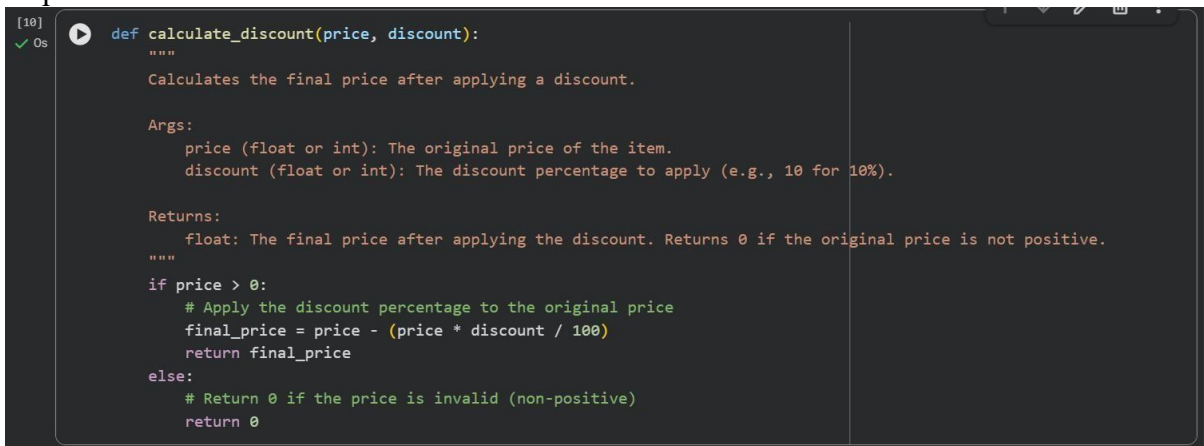
... 120
[1, 4, 9, 16]
```

Task Description -4 (Documentation – Refine Existing Documentation)

Code Used: def
calculate_discount(price, discount):
 # calculate if price > 0: # apply discount
 final_price = price - (price * discount / 100)
 return final_price
else:
 # wrong price
 return 0

Prompt:
Rewrite the documentation to improve clarity and consistency.

Output:



```
[10] def calculate_discount(price, discount):  
    """  
    Calculates the final price after applying a discount.  
  
    Args:  
        price (float or int): The original price of the item.  
        discount (float or int): The discount percentage to apply (e.g., 10 for 10%).  
  
    Returns:  
        float: The final price after applying the discount. Returns 0 if the original price is not positive.  
    """  
    if price > 0:  
        # Apply the discount percentage to the original price  
        final_price = price - (price * discount / 100)  
        return final_price  
    else:  
        # Return 0 if the price is invalid (non-positive)  
        return 0
```

Task Description -5 (Documentation – Prompt Detail Impact Study)

Code Used: def
reverse_string(text): return
text[::-1]

Prompt - 1:
Write a short documentation comment explaining what this Python
function does.

Output - 1:



```
[12] def reverse_string(text):  
    """  
    Reverses a given string.  
  
    Args:  
        text (str): The input string to be reversed.  
  
    Returns:  
        str: The reversed string.  
    """  
    return text[::-1]
```

Prompt - 2:
Write a clear and detailed documentation comment for the given Python
function.

Output - 2:

```
[13]
✓ Os
def reverse_string(text):
    """
    Reverses the given input string.

    This function takes a string as input and returns a new string
    with the characters in reverse order. It utilizes Python's string slicing
    feature to achieve this efficiently.

    Args:
        text (str): The input string that needs to be reversed.

    Returns:
        str: A new string containing the reversed characters of the input string.

    Examples:
        >>> reverse_string("hello")
        'olleh'
        >>> reverse_string("Python")
        'nohtyP'
        >>> reverse_string("12345")
        '54321'
    """
    return text[::-1]
```